

Macroeconomics

ECON 101

Where Prices Come From: The Interaction of Supply and Demand
(Chapter 3)

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Chapter Outline

3.1 The Demand Side of the Market

3.2 The Supply Side of the Market

3.3 Market Equilibrium: Putting Buyers and Sellers Together

3.4 The Effect of Demand and Supply Shifts on Equilibrium

The Demand Side of the Market

3.1 Learning Objective

Discuss the variables that influence demand.

We begin our analysis of where prices come from by investigating how buyers behave.

- We refer to this as **market demand**, the demand by all the consumers of a given good or service.

Figure 3.1 A Demand Schedule and Demand Curve (1 of 3)

Demand schedule: A table that shows the relationship between the price of a product and the quantity of the product demanded.

Demand curve: A curve that shows the relationship between the price of a product and the quantity of the product demanded.

Demand Schedule	
Price (dollars per pair of shoes)	Quantity (millions of pairs of shoes per week)
\$150	8
125	9
100	10
75	11
50	12

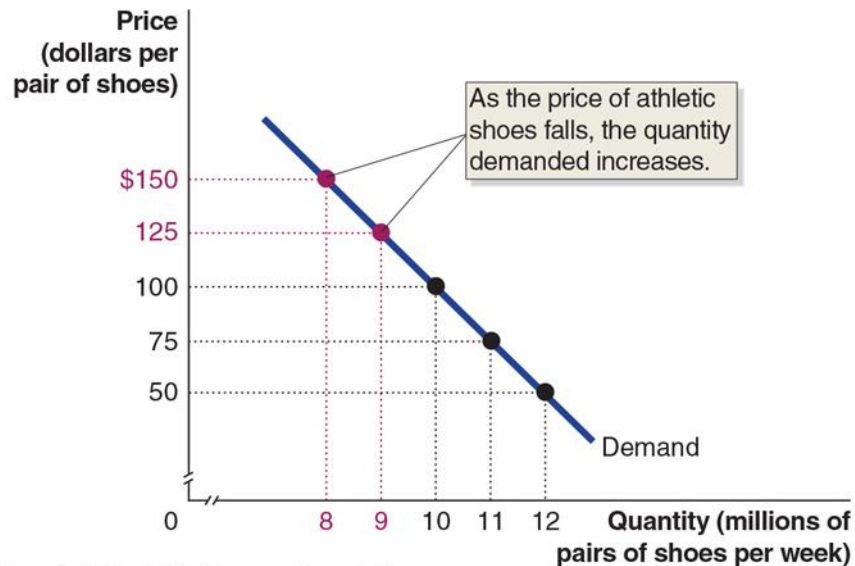


Figure 3.1 A Demand Schedule and Demand Curve (2 of 3)

When drawing the demand curve, we assume **ceteris paribus**.

Ceteris paribus (“all else equal”) condition: The requirement that when analyzing the relationship between two variables—such as price and quantity demanded—other variables must be held constant.

Demand Schedule	
Price (dollars per pair of shoes)	Quantity (millions of pairs of shoes per week)
\$150	8
125	9
100	10
75	11
50	12

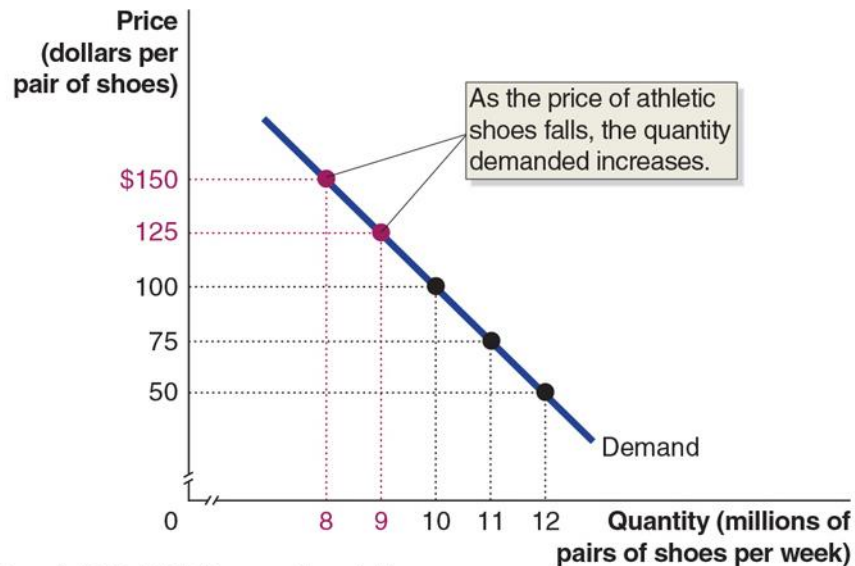
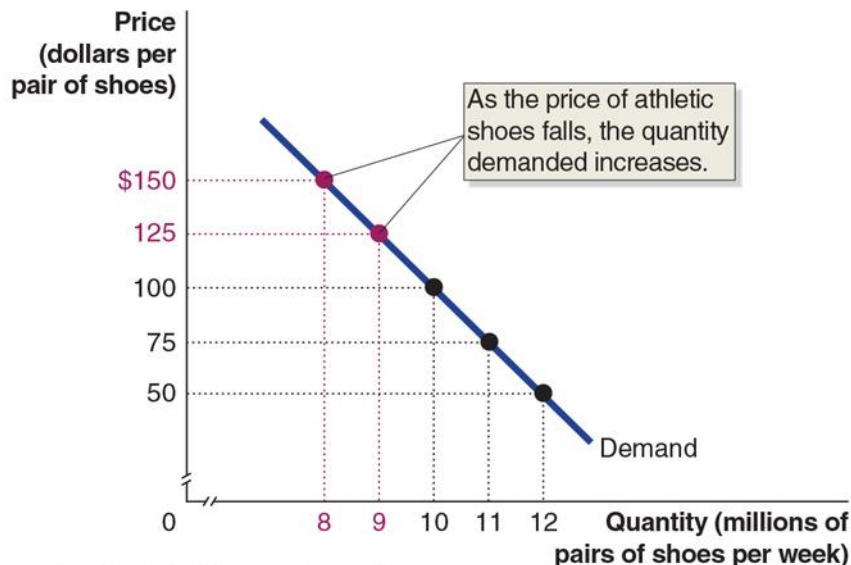


Figure 3.1 A Demand Schedule and Demand Curve (3 of 3)

Quantity demanded: The amount of a good or service that a consumer is willing and able to purchase at a given price.

Law of demand: A rule that states that, holding everything else constant, when the price of a product falls, the quantity demanded of the product will increase, and when the price of a product rises, the quantity demanded of the product will decrease.

Demand Schedule	
Price (dollars per pair of shoes)	Quantity (millions of pairs of shoes per week)
\$150	8
125	9
100	10
75	11
50	12



What Explains the Law of Demand?

When the price of a good falls, two effects take place:

1. Consumers **substitute** toward the good whose price has fallen.
2. Consumers have more purchasing power, which is like an increase in **income**.

We call these the **substitution effect** and the **income effect**:

- **Substitution effect:** The change in the quantity demanded of a good that results from a change in price, making the good more or less expensive relative to other goods, holding constant the effect of the price change on consumer purchasing power.
- **Income effect:** The change in the quantity demanded of a good that results from the effect of a change in the good's price on a consumers' purchasing power, holding all other factors constant.

Figure 3.2 Shifting the Demand Curve

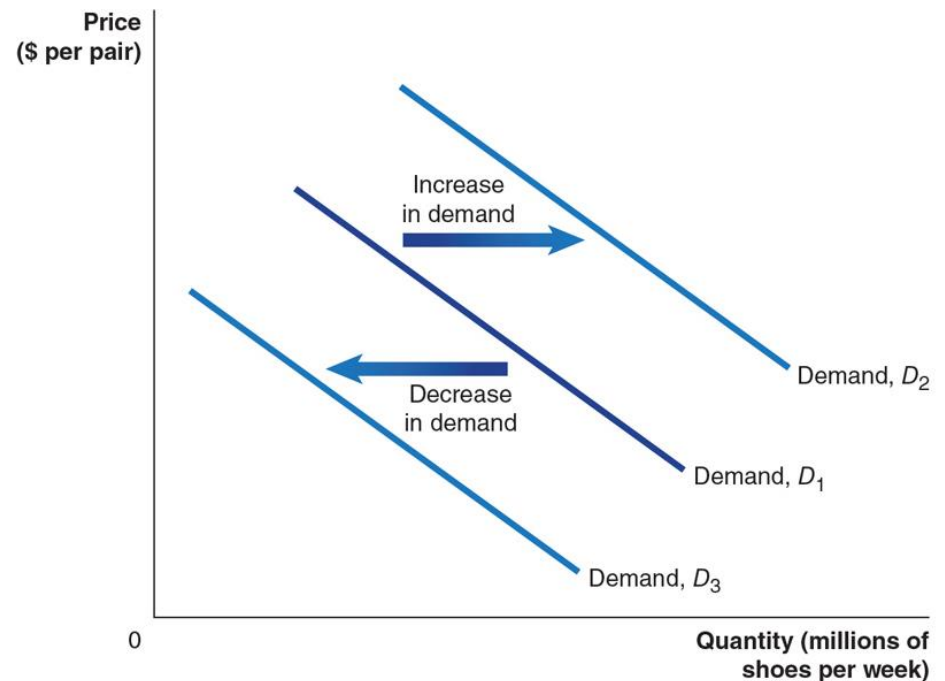
(1 of 2)

A change in something other than price that affects demand causes the entire demand curve to shift.

is a **decrease in demand**.

A shift to the right (D_1 to D_2) is an **increase in demand**.

A shift to the left (D_1 to D_3) is a **decrease in demand**.



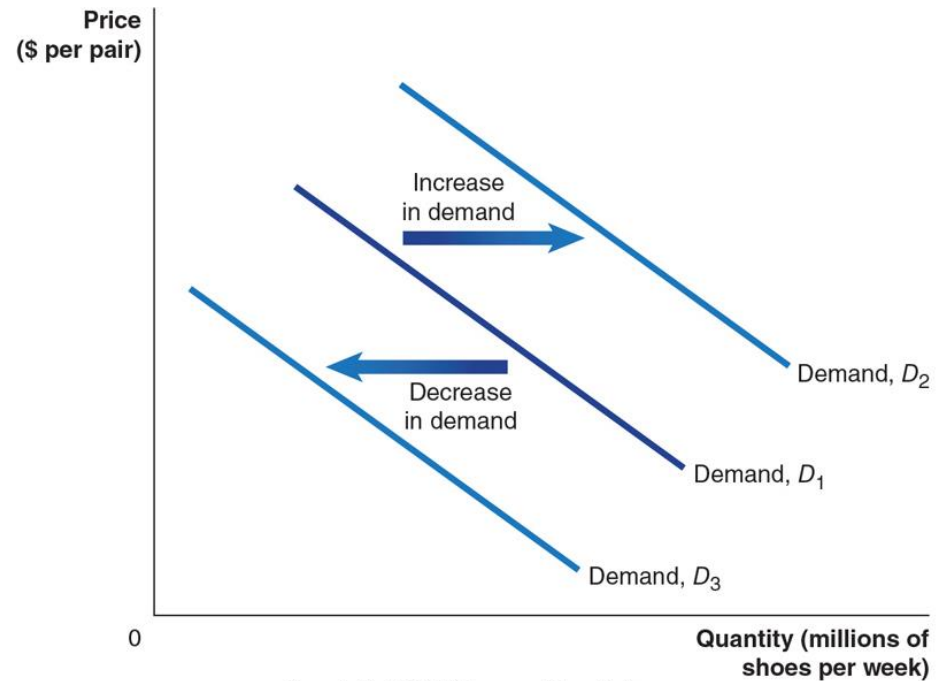
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Figure 3.2 Shifting the Demand Curve

(2 of 2)

As the demand curve shifts, the quantity demanded will change, **even if the price doesn't change.**

The quantity demanded changes at every possible price.



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Variables That Shift Market Demand

- **Income of consumers**

- Increase in income increases demand if the product is **normal**, decreases demand if the product is **inferior**.

- **Prices of related goods**

- Increase in price of related good increases demand if products are **substitutes**, decreases demand if products are **complements**

- **Tastes**

- **Population and demographics**

- **Expectations**

We will discuss how each of these affects demand.

Changes in Income of Consumers

Normal goods: Goods for which the demand increases as income rises and decreases as income falls.

Examples: New clothes
 Restaurant meals
 Vacations

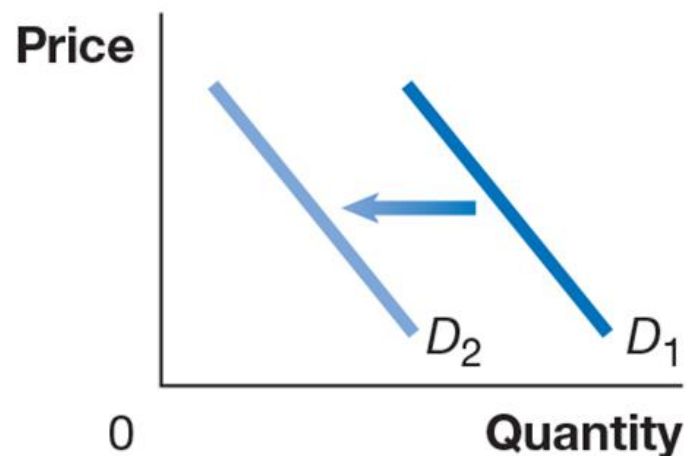
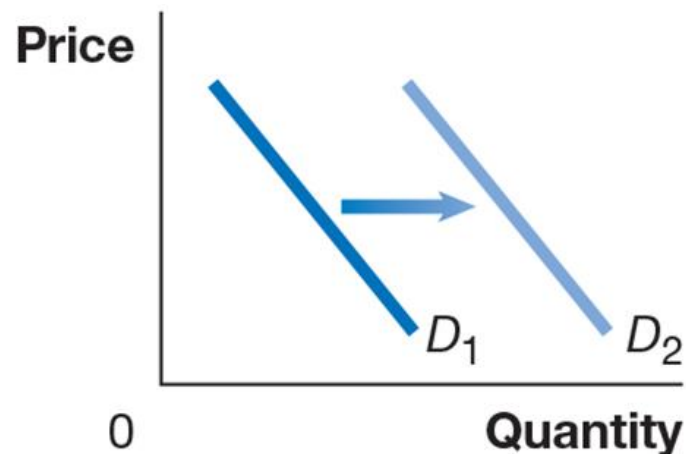
Inferior goods: Goods for which the demand increases as income falls and decreases as income rises.

Examples: Second-hand clothes
 Instant noodles

Effects of Changes in Income

An increase in income would increase the demand for new clothes, **ceteris paribus**.

However, the same increase in income would likely **decrease** the demand for second-hand clothes.



Apply the Concept: The Transformation of Lobster from Inferior to Normal Good

Describing a good as normal or inferior only tells us about how consumers behave.

Do people buy more or less when their incomes rise?

In some parts of the Maritimes, lobster is an inferior good – people eat less of it when their incomes rise.

In Toronto or Calgary, lobster is a normal good – people eat more when their incomes rise.

Goods can be normal in one place and inferior in another.

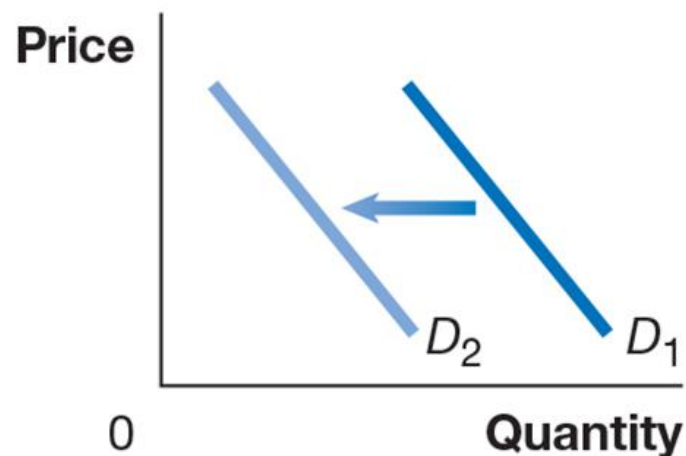
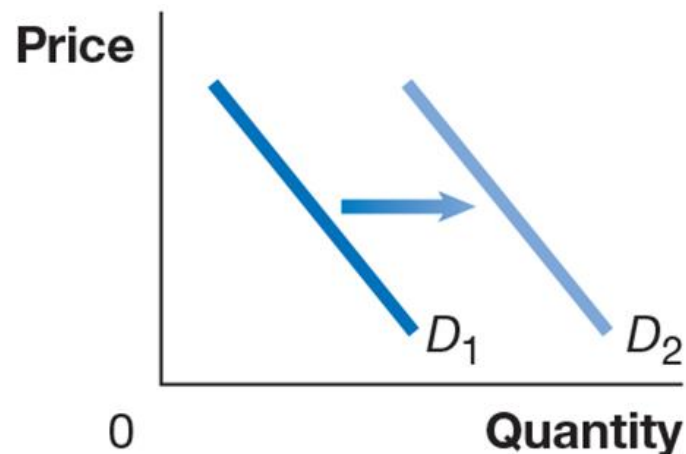
Despite its high cost in many restaurants, lobster can still be an inferior good.



Effects of Changes in the Price of Related Goods

An increase in the price of a Big Mac would increase the demand for Whoppers.

However, the same increase in the price of a Big Mac would decrease the demand for McDonald's fries.

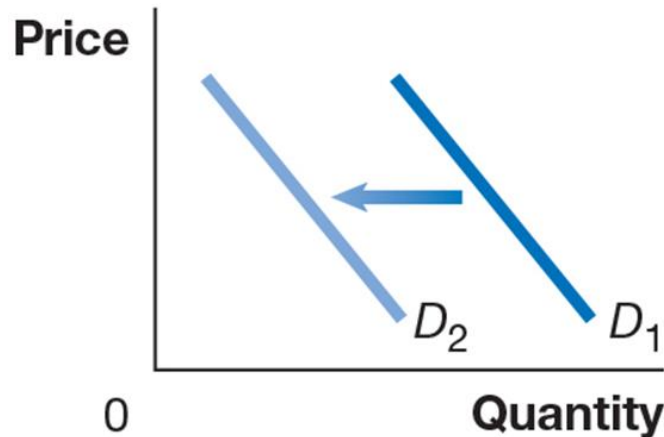


Changes in Tastes

Tastes

If consumers' tastes change, they may buy more or less of the product.

Example: If consumers become more concerned about eating healthy, they might decrease their demand for fast food.

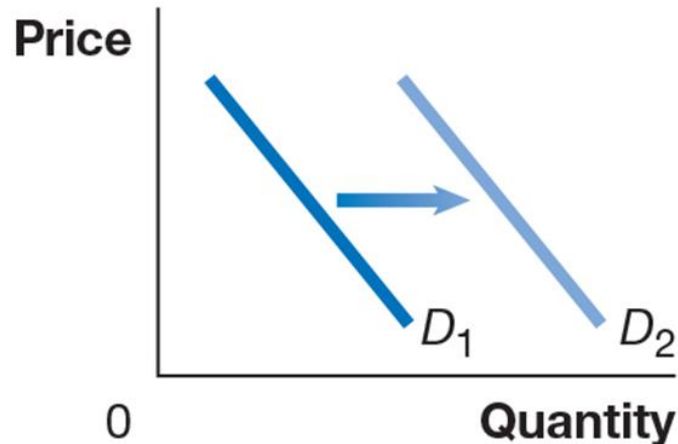


Changes in Population and Demographics

Demographics: The characteristics of a population with respect to age, race, and gender.

Increases in the number of people buying something will increase the amount demanded.

Example: An increase in the elderly population increases the demand for medical care.



Changes in Expectations About Future Prices

Consumers decide **which** products to buy and also **when** to buy them.

- Future products are **substitutes** for current products.
- An expected **increase** in the price tomorrow **increases demand today**.
- An expected **decrease** in the price tomorrow **decreases demand today**.

Example: If you found out the price of gasoline would go up tomorrow, you would increase your demand today.

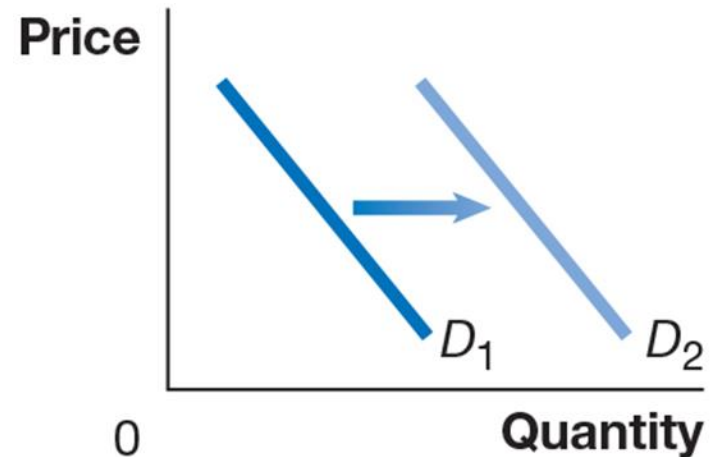


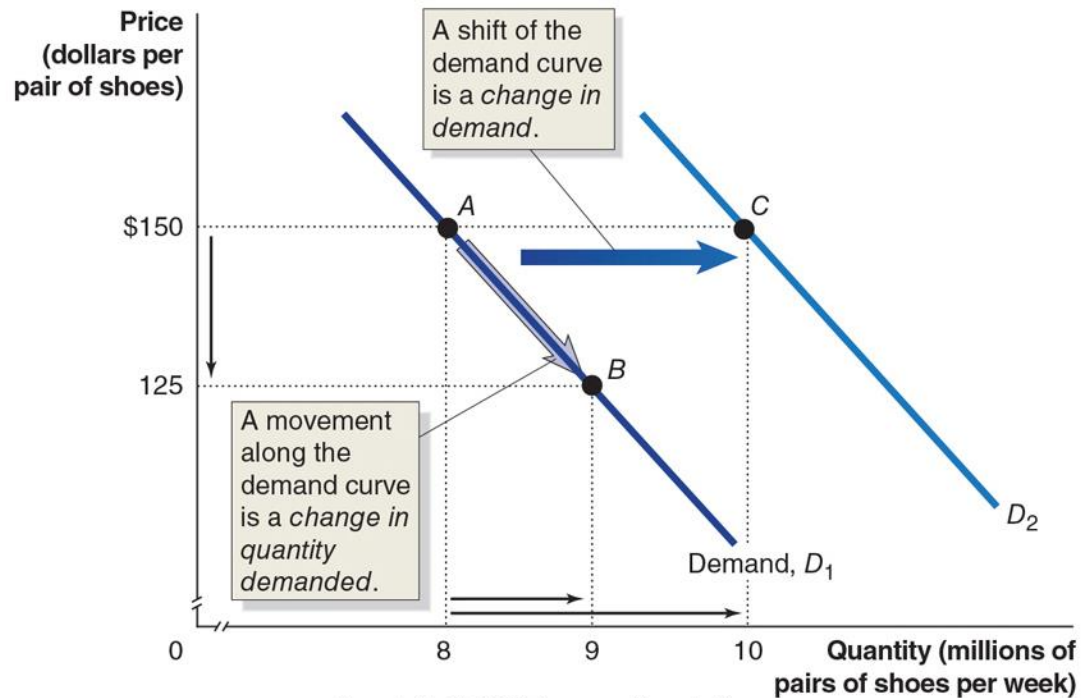
Figure 3.3 A Change in Demand vs a Change in Quantity

A change in the price of the product being examined causes a movement along the demand curve.

- This is a **change in quantity demanded**.

Any other change affecting demand causes the entire demand curve to shift.

- This is a **change in demand**.



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The Supply Side of the Market

3.2 Learning Objective

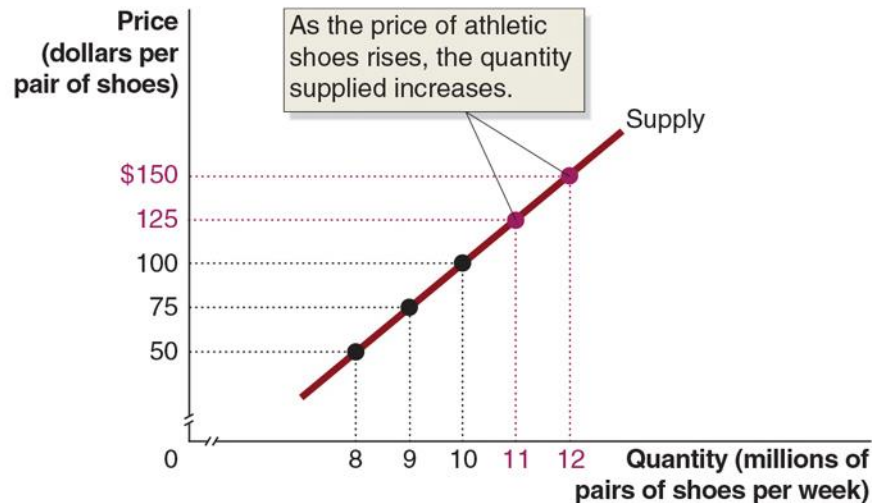
Discuss the variables that influence supply.

There are some similarities, and some important differences, between the demand and supply sides of the market.

In this section, we examine the **market supply**, i.e., the decisions of (generally) firms about how much of a product to provide at various prices.

Figure 3.4 A Supply Schedule and Supply Curve (1 of 2)

Supply Schedule	
Price (dollars per pair of shoes)	Quantity (millions of pairs of shoes per week)
\$150	12
125	11
100	10
75	9
50	8



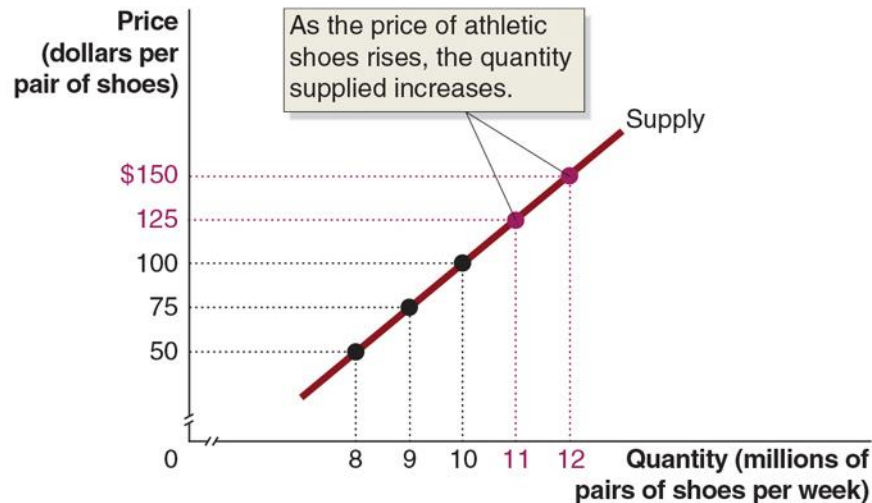
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Supply schedule: A table that shows the relationship between the price of a product and the quantity of the product supplied.

Supply curve: A curve that shows the relationship between the price of a product and the quantity of the product supplied.

Figure 3.4 A Supply Schedule and Supply Curve (2 of 2)

Supply Schedule	
Price (dollars per pair of shoes)	Quantity (millions of pairs of shoes per week)
\$150	12
125	11
100	10
75	9
50	8



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Quantity supplied: The amount of a good or service that a firm is willing and able to supply at a given price.

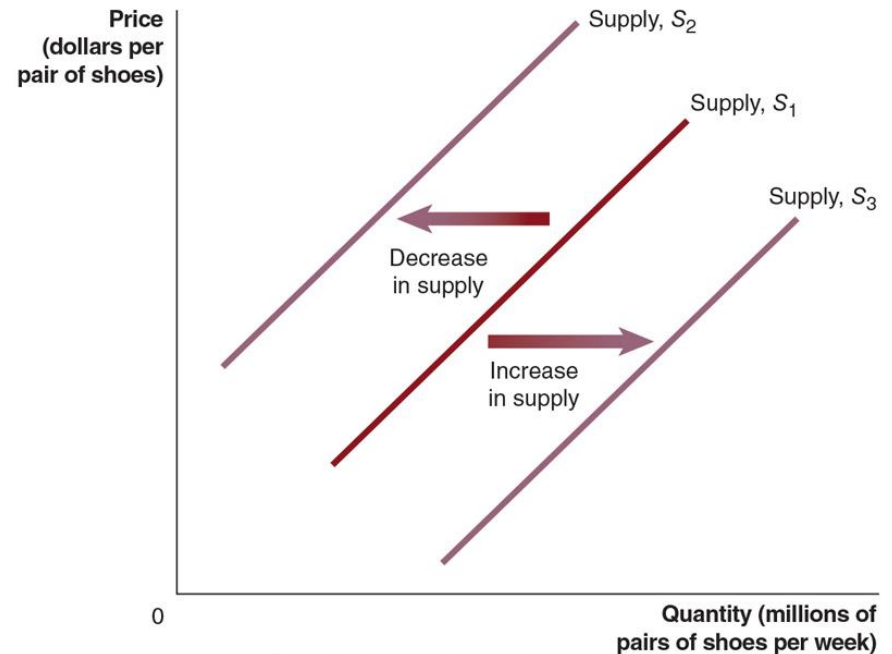
Law of supply: A rule that states that, holding everything else constant, increases in price cause increases in the quantity supplied, and decreases in price cause decreases in the quantity supplied.

Figure 3.5 Shifting the Supply Curve

(1 of 2)

A change in something other than price that affects supply causes the entire supply curve to shift.

- A shift to the right (S_1 to S_3) is an **increase in supply**.
- A shift to the left (S_1 to S_2) is a **decrease in supply**.



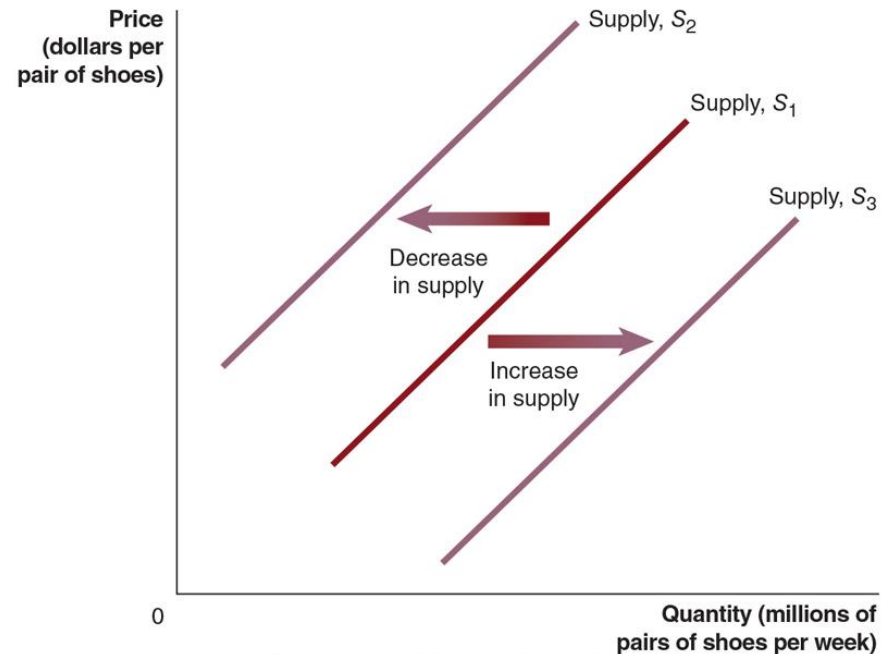
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Figure 3.5 Shifting the Supply Curve

(2 of 2)

As the supply curve shifts, the quantity supplied will change, **even if the price doesn't change.**

The quantity supplied changes at every possible price.



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What Variables Shift Market Supply?

Prices of inputs

Technological change

Prices of substitutes in production

Number of firms in the market

Expected future prices

We will discuss how each of these affect supply.

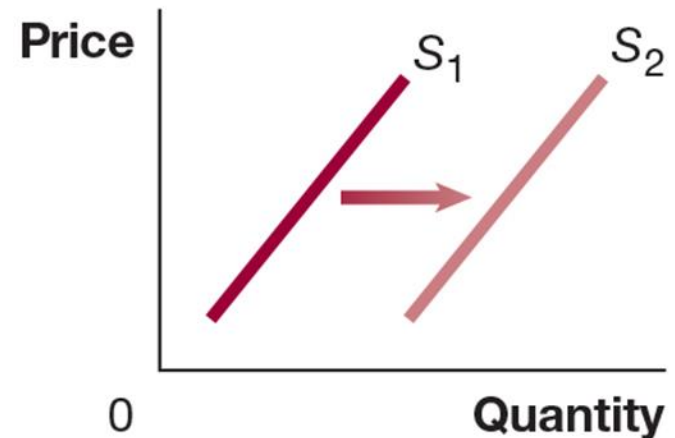
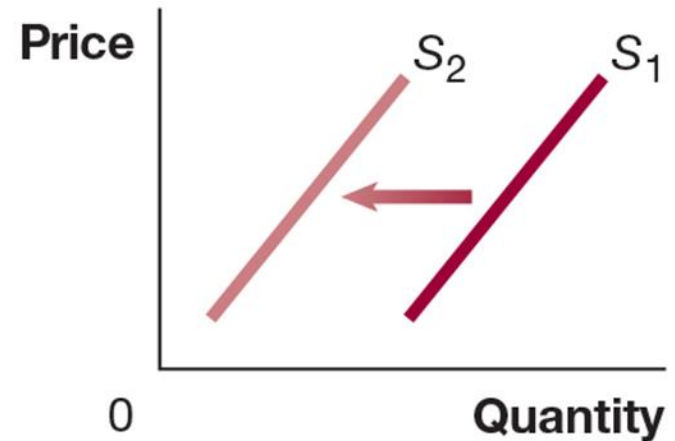
Change in Prices of Inputs

Inputs are anything used in the production of a good or service.

For an athletic shoe, inputs include the rubber, plastic, and labor.

An **increase in the price of an input** decreases the profitability of selling the good, causing a **decrease in supply**.

A **decrease in the price of an input** increases the profitability of selling the good, causing an **increase in supply**.

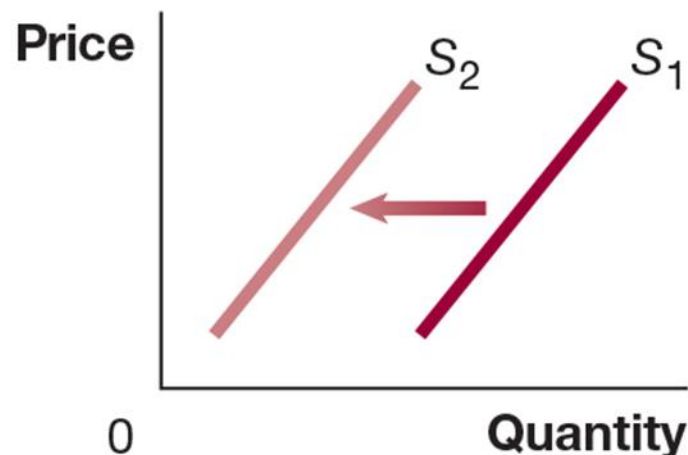
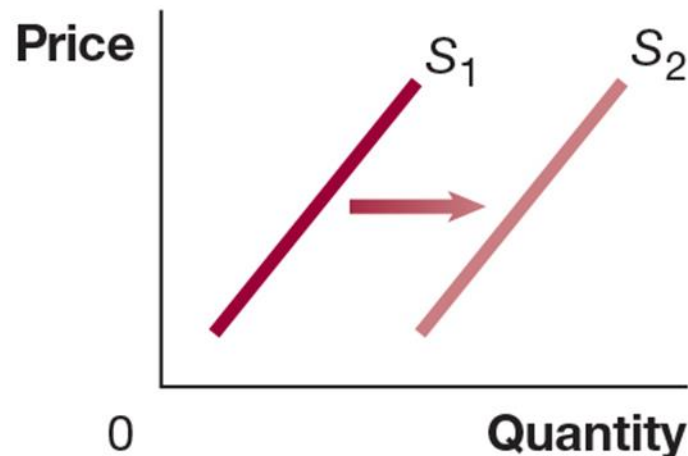


Technological Change

A firm may experience a positive or negative change in its ability to produce a given level of output with a given quantity of inputs. We call this a **technological change**.

Examples:

- A new, more efficient way of producing shoes would increase their supply.
- Governmental restrictions on how much workers are allowed to work might decrease the supply of athletic shoes.



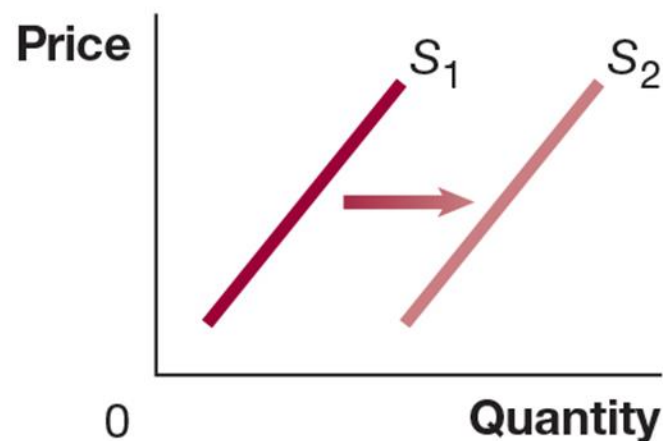
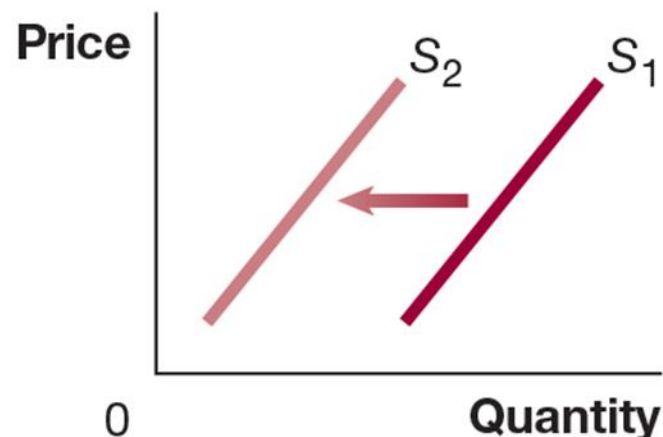
Prices of Substitutes in Production

Many firms can produce and sell alternative products: **substitutes in production**.

Example: An Ontario farmer can plant either corn or soybeans. If the price of soybeans rises, that farmer will plant (supply) less corn.

Sometimes, two products are necessarily produced together: **complements in production**

Example: Cattle provide both beef and leather. An increase in the price of beef encourages more cattle farming, which increases the supply of leather.



Number of Firms in the Market and Expected Future Prices

More firms in the market will result in more product available at a given price (greater supply).

Fewer firms → supply decreases.

If a firm anticipates that the price of its product will be higher in the future, it might decrease its supply today in order to increase it in the future.

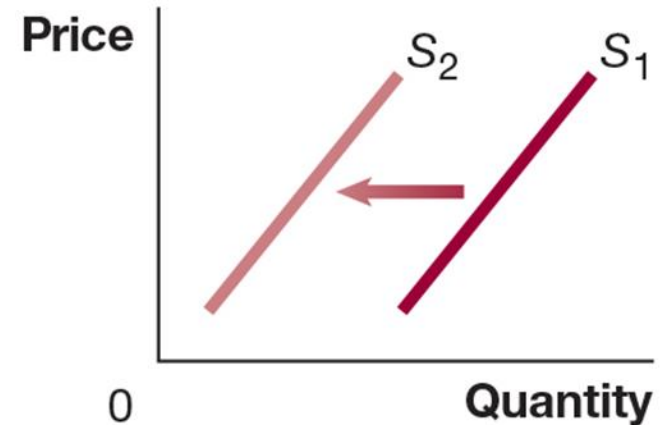
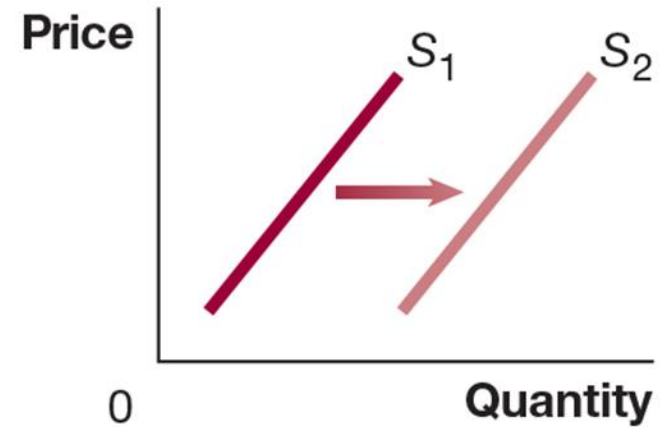


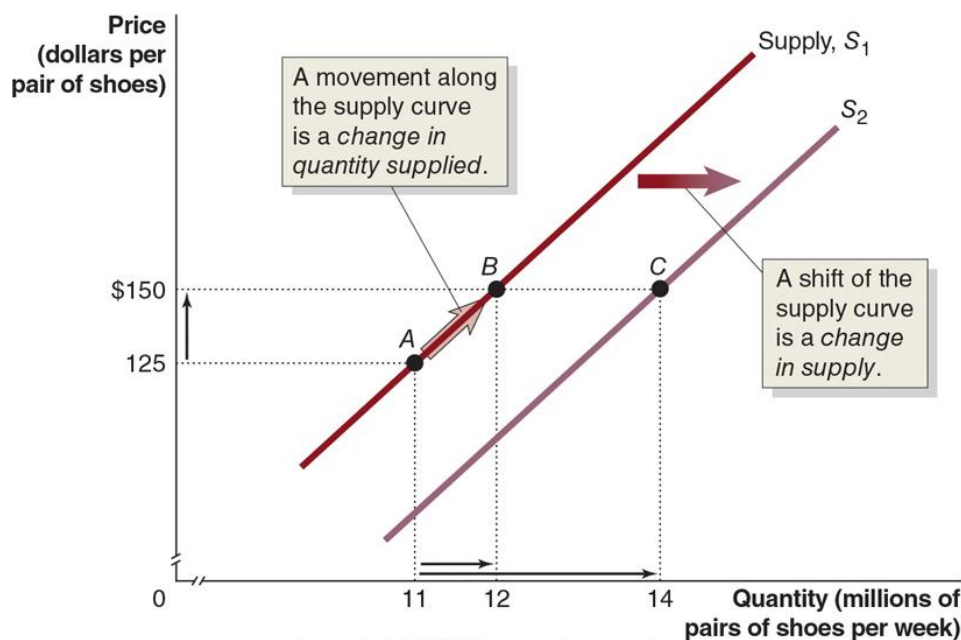
Figure 3.6 A Change in Supply Versus a Change in Quantity Supplied

A change in the price of the product being examined causes a movement along the supply curve.

- This is a **change in quantity supplied**.

Any other change affecting supply causes the entire supply curve to shift.

- This is a **change in supply**.



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Market Equilibrium: Putting Buyers and Seller Together

3.3 Learning Objective

Use a graph to illustrate market equilibrium.

Market equilibrium is a situation in which quantity demanded equals quantity supplied.

Recall that markets with many buyers and sellers are perfectly competitive markets; a market equilibrium in one of these markets is called a **competitive market equilibrium**.

There are ~35 firms selling athletic shoes; we will assume this is enough to generate competitive behaviour in the market for athletic shoes.

Figure 3.7 Market Equilibrium

At a price of \$100,

- consumers want to buy 10 million pairs of shoes per week, and
- producers want to sell 10 million pairs of shoes per week.

We say the **equilibrium price** in this market is \$100, and the **equilibrium quantity** is 10 million pairs of shoes per week.

Since buyers and sellers want to trade the same quantity at the price of \$100, we do not expect the price to change.

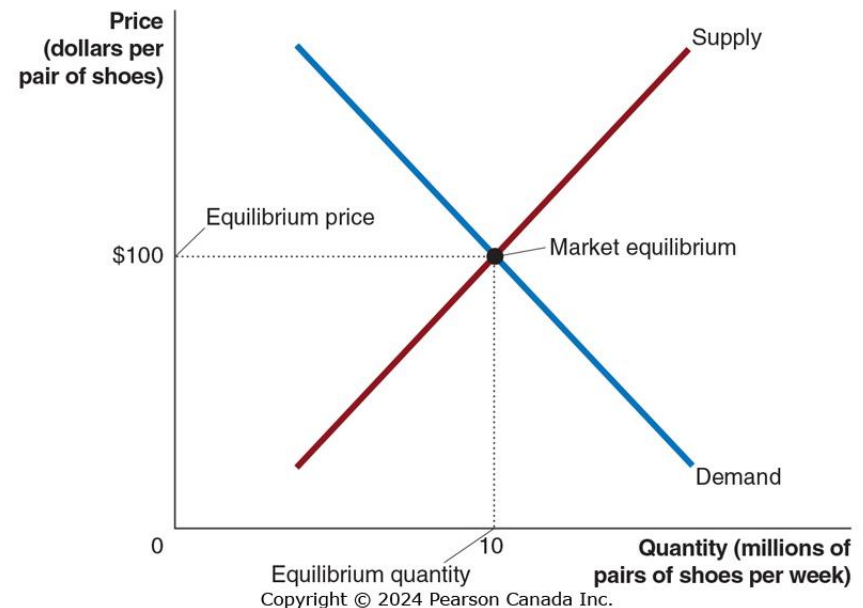


Figure 3.8 The Effect of Surpluses and Shortages on the Market Price (1 of 2)

What if the price were \$125 instead?

At a price of \$125,

- consumers want to buy 9 million pairs of shoes, while
- producers want to sell 11 million pairs.

This gives a **surplus** of 2 million pairs; a situation in which quantity supplied is greater than quantity demanded.

Prediction: Sellers will compete among themselves, driving the price down.

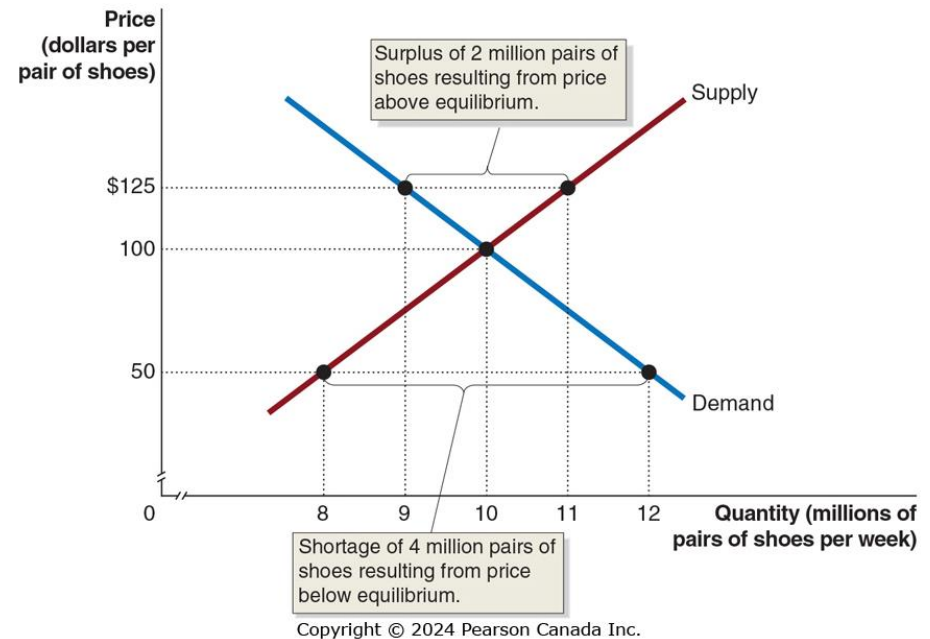


Figure 3.8 The Effect of Surpluses and Shortages on the Market Price (2 of 2)

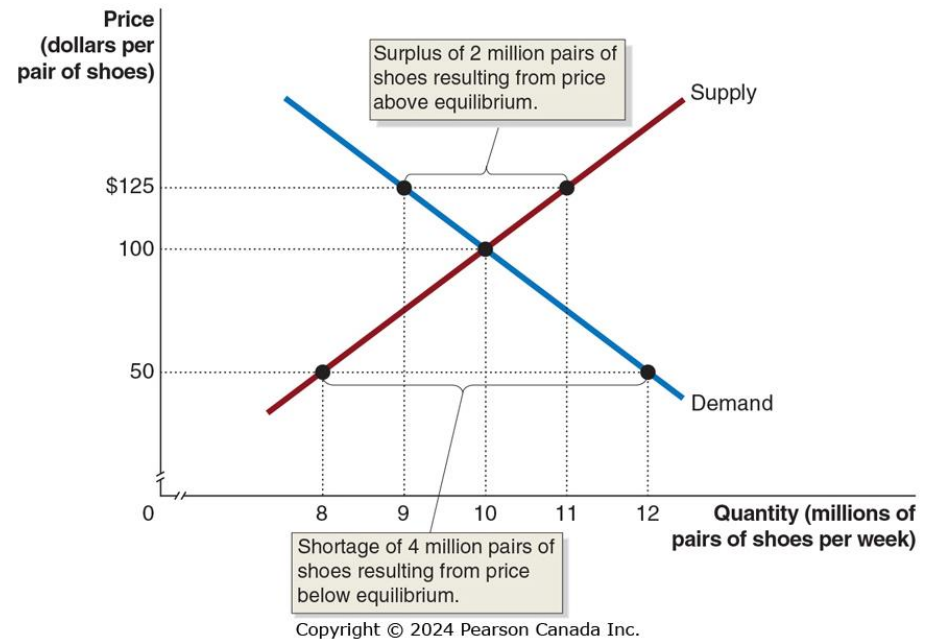
Now what if the price were \$50?

At a price of \$50,

- consumers want to buy 12 million pairs of shoes, while
- producers want to sell 8 million.

This gives a **shortage** of 4 million pairs of shoes; a situation in which quantity demanded is greater than quantity supplied.

Prediction: Sellers will realize they can increase the price and still sell as many pairs of shoes, so the price will rise.



Demand and Supply Both Count

Price is determined by the **interaction** of buyers and sellers.

Neither group can dictate price in a competitive market (i.e. one with many buyers and sellers).

However, **changes in supply and/or demand** will affect the price and quantity traded.

The Effect of Demand and Supply Shifts on Equilibrium

3.4 Learning Objective

Use demand and supply graphs to predict changes in prices and quantities.

Predictions about price and quantity in our model require us to know demand and supply curves.

- Typically, we know price and quantity but **do not know** the curves that generate them.

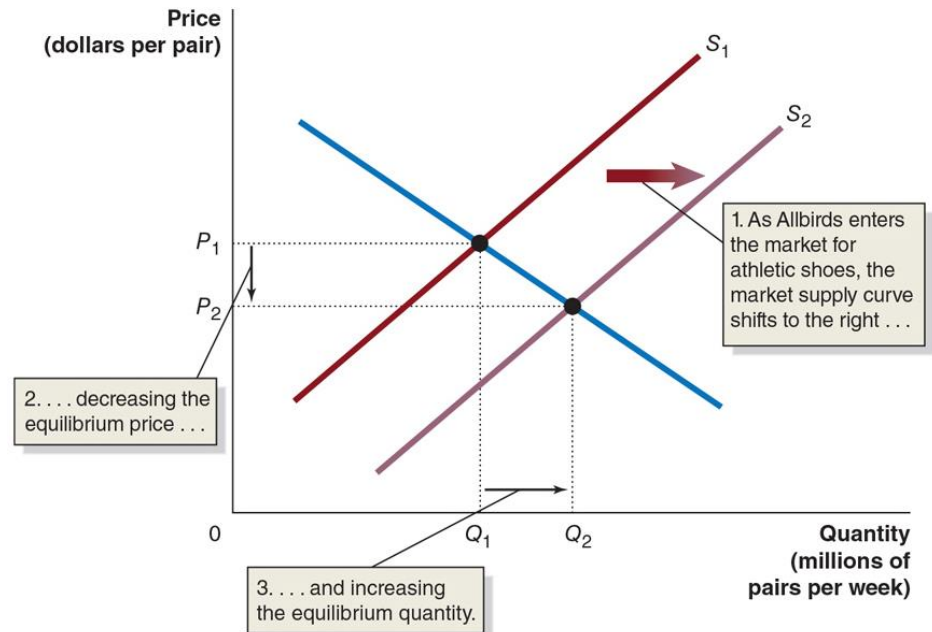
The power of the demand and supply model is in its ability to predict *directional changes* in price and quantity traded.

Figure 3.9 The Effect of an Increase in Supply on Equilibrium (1 of 2)

The graph shows the market for athletic shoes as a new shoemaker, Allbirds, enters the market.

When Allbirds enters, more shoes are supplied at any given price—an increase in supply from S_1 to S_2 .

- Equilibrium price falls from P_1 to P_2 .
- Equilibrium quantity rises Q_1 to Q_2 .



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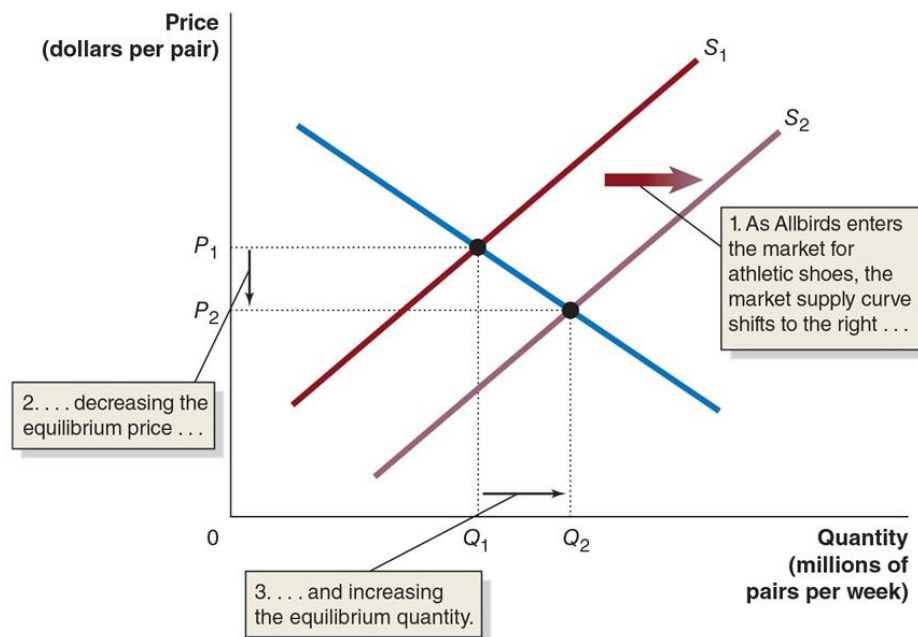
Figure 3.9 The Effect of an Increase in Supply on Equilibrium (2 of 2)

By how much will price fall?

By how much will quantity rise?

We cannot say, without knowing more information.

For now, we can only predict that price will fall and quantity will rise.



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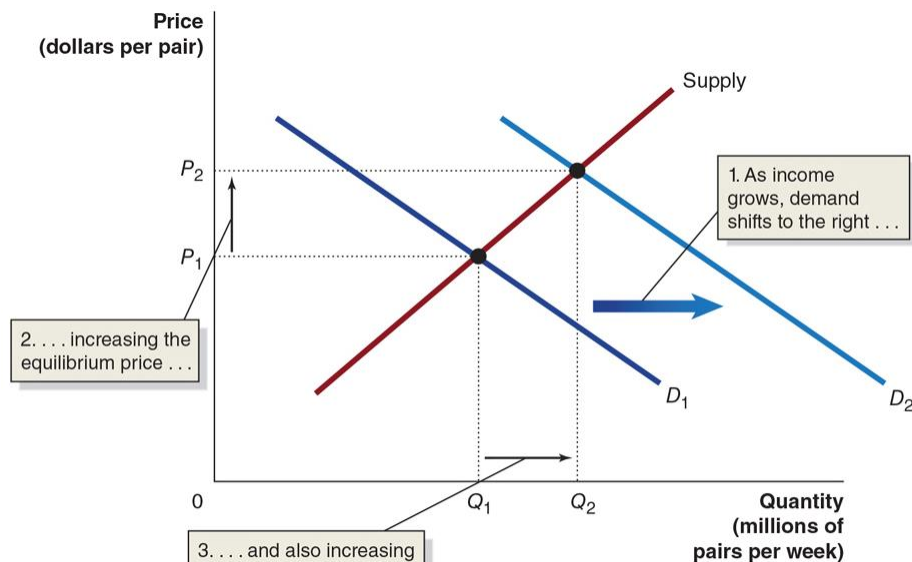
Figure 3.10 The Effect of an Increase in Demand on Equilibrium

Suppose incomes increase.
What happens to the equilibrium
in the athletic shoe market?

Athletic shoes are a **normal good**, so as income rises, demand shifts to the right

(D_1 to D_2).

- Equilibrium price rises
(P_1 to P_2).
- Equilibrium quantity rises
(Q_1 to Q_2).



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Table 3.3 How Shifts in Demand and Supply Affect Equilibrium Price (P) and Quantity (Q) (1 of 2)

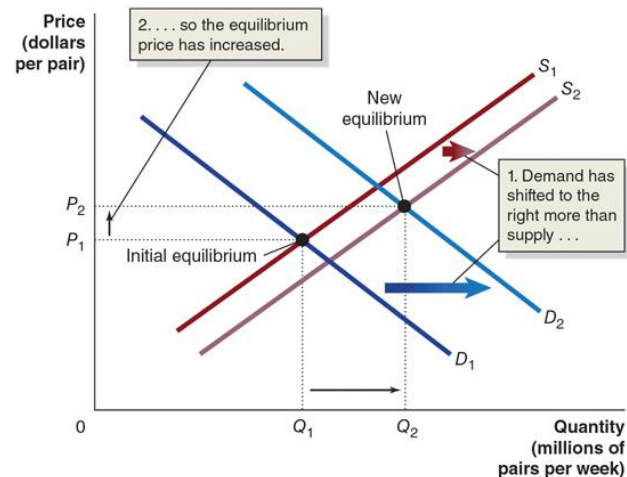
	Supply Curve Unchanged	Supply Curve Shifts to the Right	Supply Curve Shifts to the Left
Demand Curve Unchanged	Q unchanged P unchanged	Q increases P decreases	Q decreases P increases
Demand Curve Shifts to the Right	Q increases P increases	Q increases P increases or decreases	Q increases or decreases P increases
Demand Curve Shifts to the Left	Q decreases P decreases	Q increases or decreases P decreases	Q decreases P increases or decreases

Figure 3.11 Shifts in Demand and Supply Over Time (1 of 3)

Over time, it is likely that **both** demand and supply will change.

For example, as new firms enter the market for athletic shoes **and** incomes increase, we expect:

- The supply curve will shift to the right, **and**
- The demand curve will shift to the right.



(a) Demand shifting more than supply
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Figure 3.11 Shifts in Demand and Supply Over Time (2 of 3)

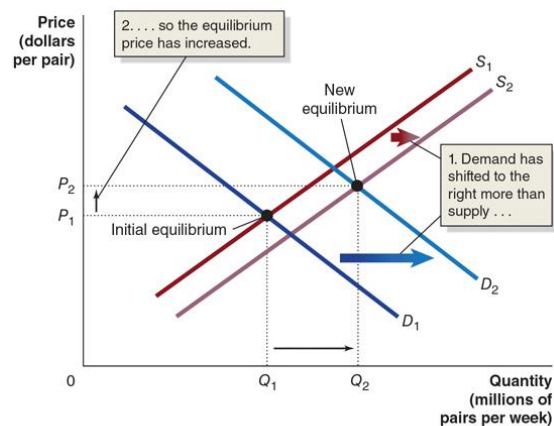
What does our model predict?

$S \uparrow \rightarrow (P \downarrow \text{ and } Q \uparrow)$

$D \uparrow \rightarrow (P \uparrow \text{ and } Q \uparrow)$

So we can be sure equilibrium quantity will rise, but the effect on equilibrium price is not clear.

This panel shows demand shifting more than supply: equilibrium price and quantity both rise.



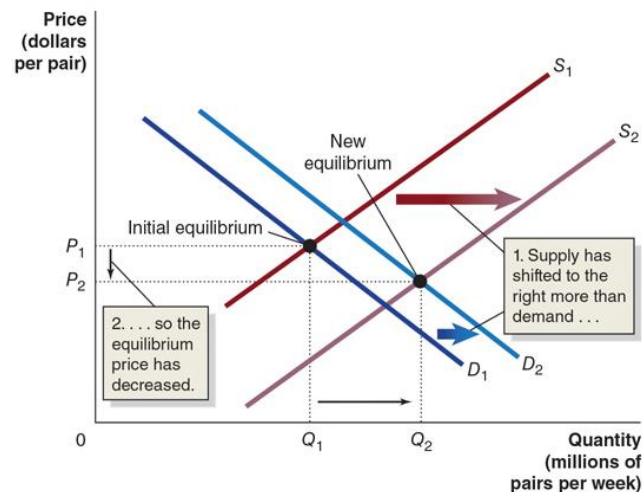
(a) Demand shifting more than supply
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Figure 3.11 Shifts in Demand and Supply Over Time (3 of 3)

This panel shows supply shifting more than demand: quantity rises, but equilibrium price falls.

Without knowing the relative size of the changes, the effect on equilibrium price is ambiguous.

It is possible, but unlikely, that the equilibrium price will remain unchanged.



(b) Supply shifting more than demand
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Table 3.3 How Shifts in Demand and Supply Affect Equilibrium Price (P) and Quantity (Q) (2 of 2)

	Supply Curve Unchanged	Supply Curve Shifts to the Right	Supply Curve Shifts to the Left
Demand Curve Unchanged	Q unchanged P unchanged	Q increases P decreases	Q decreases P increases
Demand Curve Shifts to the Right	Q increases P increases	Q increases P increases or decreases	Q increases or decreases P increases
Demand Curve Shifts to the Left	Q decreases P decreases	Q increases or decreases P decreases	Q decreases P increases or decreases

The cell in red is the example that we just did.

Shifts of a Curve vs. Movements Along a Curve

Suppose an increase in supply occurs. We now know:

- Equilibrium quantity will increase, and
- Equilibrium price will decrease

It is tempting to believe the decrease in price will cause an increase in demand. But this is incorrect.

- The decrease in price will cause a movement along with demand curve, but *not* an increase in demand.
- Why? The demand curve *already* describes how much of the good consumers want to buy, *at any given price*.
- When the price change occurs, we just look at the demand curve to see what happens to how much consumers want to buy.